

# **Report**

Of

## **Development of a Frontend-Based E-Commerce Shopping Website “ShopSphere” for INFUST PVT LTD**

Submitted

To

**School of Computer Science and Engineering  
IILM University, Greater Noida, U.P.**

In partial fulfilment of the requirement of degree of  
**B.Tech CSE**



By

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**Batch: 2026-27**

## **CANDIDATE'S DECLARATION**

I, Luv Singh Chauhan, do hereby declare that the internship report titled “Development of a Frontend-Based E-Commerce Shopping Website ‘ShopSphere’ for INFUST PVT LTD” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm that this report is my own work and has not been submitted to any other institute for any degree/diploma requirement, and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature

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## **ACKNOWLEDGEMENT**

I am using this opportunity to express my gratitude to everyone who supported me throughout my internship at INFUST PVT LTD. I am thankful for their aspiring guidance, invaluable constructive criticism, and friendly advice during this work. I would like to sincerely thank Er. Rakesh Kumar, Managing Director, INFUST PVT LTD, for giving me the opportunity to work as a Web Developer Intern and for his constant support and mentorship throughout the internship period from 10 July 2026 to 14 August 2026.

I would also like to thank the faculty of the School of Computer Science and Engineering, IILM University, Greater Noida, for their continuous guidance and encouragement, which helped me apply my academic knowledge to a real, industry-oriented project. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents and friends for their blessings and constant encouragement throughout this internship.

Signature

Student Name: Luv Singh Chauhan

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# INTERNSHIP COMPLETION CERTIFICATE



*Certificate issued by INFUST PVT LTD, Mumbai, to Luv Singh Chauhan in recognition of the successful completion of his internship as a Web Developer.*

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## **CHAPTER 4: PROJECT DESCRIPTION**

### **4.1 Introduction**

This report documents the work carried out during my internship as a Web Developer at INFUST PVT LTD, Mumbai, from 10 July 2026 to 14 August 2026. During this internship, I was assigned to design and develop the frontend of a full-featured e-commerce shopping website named “ShopSphere” for INFUST PVT LTD. The internship provided hands-on, industry-level exposure to modern frontend web development practices, including client-side state management, responsive UI design, and building an application that behaves like a production-grade online shopping platform without relying on a backend server.

The project involved building every core module that a real shopping website requires — product browsing, search and filtering, a shopping cart, a wishlist, a multi-step checkout flow, an order-tracking system, a user profile dashboard, and an administrative panel for managing the product catalogue — entirely using frontend technologies. This report explains the organization I interned with, the problem the project addresses, its objectives and scope, the technologies used, the system architecture, the methodology followed during development, and the outcomes achieved.

### **4.2 Organization Profile**

INFUST PVT LTD is a Mumbai-based technology company that undertakes web and software development projects and provides internship training to students in real-world project environments. The organization focuses on practical, project-based learning, pairing interns with a live deliverable rather than purely theoretical assignments. My internship was supervised by Er. Rakesh Kumar, Managing Director, INFUST PVT LTD, who guided the requirements, reviewed progress, and provided feedback throughout the internship duration.

### **4.3 Problem Statement**

Many students and early-career developers learn isolated web development concepts — HTML, CSS, and JavaScript — without ever assembling them into a single, cohesive, functioning application. Similarly, small businesses often need a working

demonstration of an online storefront before committing to the cost and complexity of a full backend, database, and payment gateway integration. The problem addressed by this project was: how can a complete, realistic online shopping experience — covering browsing, searching, cart management, checkout, order tracking, and product administration — be built using only frontend technologies, while still persisting user data reliably across browser sessions and behaving exactly as a backend-connected application would from the user's point of view.

#### **4.4 Project Objectives**

- To design and build a fully functional, frontend-only e-commerce website for INFUST PVT LTD without using any backend server, database, or real payment gateway.
- To implement core shopping features — product listing, search, category and price filtering, sorting, and a detailed product view — driven entirely by client-side JavaScript.
- To build a persistent shopping cart and wishlist system so that a user's selections are retained even after refreshing or closing the browser.
- To create a realistic, multi-step checkout flow covering address entry, delivery option selection, a simulated payment step, order review, and order confirmation.
- To provide a user profile section and an order-history/order-tracking view, along with an administrative panel that allows products to be added, edited, or removed and immediately reflected on the storefront.
- To ensure the entire interface is responsive, accessible, and visually consistent across desktop, tablet, and mobile screen sizes.

#### **4.5 Scope of the Project**

The scope of the project was limited strictly to the frontend layer of the application. All product data, user profile information, cart and wishlist contents, addresses, orders, and admin-side product edits are created, modified, and stored using the browser's localStorage, which simulates persistent data storage without a real backend. The project covers the Home page, Shop/Product-listing page with filters and sorting, individual Product Detail pages, Cart, Wishlist, a multi-step Checkout flow, a Profile

dashboard, an Orders/Order-tracking section, and an Admin panel for catalogue management. Real payment processing, user authentication against a server, and persistent multi-device data synchronization were explicitly outside the scope of this internship project, since these require backend infrastructure.

#### **4.6 Technologies and Tools Used**

- HTML5 – for the semantic structure of the single-page application.
- CSS3 / Tailwind CSS – for a modern, responsive, utility-first visual design.
- JavaScript (ES6+) – for all application logic, including a lightweight client-side router, state management, and rendering functions.
- Browser localStorage API – to persist the product catalogue, cart, wishlist, user profile, addresses, orders, and reviews across browser sessions.
- Google Fonts – for consistent, brand-appropriate typography across the interface.
- Visual Studio Code – as the code editor used for development, and the Chrome browser's DevTools for debugging and responsive testing.

#### **4.7 System Architecture**

ShopSphere follows a single-page application (SPA) architecture built entirely in the browser. A central JavaScript router function reads the requested route (such as the home page, the shop listing, an individual product, the cart, checkout, profile, orders, or the admin panel) and calls the matching render function, which regenerates the relevant portion of the page. Product, cart, wishlist, order, and profile data are held in a JavaScript state layer at runtime and are synchronized to and from localStorage through dedicated persistence functions, so the same state is restored automatically whenever the page is reloaded. Because there is no backend, the admin panel writes product changes directly into this shared state, and the storefront reads from the same state, so any change made in the admin section is reflected immediately on the shop and product pages.

#### **4.8 Methodology**

The project was developed iteratively over the course of the internship. The first phase focused on planning the page structure, the product data model, and the overall visual design of the storefront under the INFUST PVT LTD brand. The second phase involved



building the static layout — header, navigation, footer, and product cards — followed by wiring up client-side routing between pages. The third phase implemented interactive functionality: search, category/price/rating filters, sorting, the cart and wishlist systems with live badge counters, and toast notifications for user actions. The fourth phase built the multi-step checkout flow (address, delivery method, payment simulation, order review, and confirmation) and the order-tracking view. The final phase added the profile dashboard and the admin panel for product management, followed by responsive-design refinement and end-to-end testing of the complete user journey, from browsing a product to placing an order and viewing it in order history.

#### **4.9 Expected Outcomes**

- A fully working shopping website prototype for INFUST PVT LTD that demonstrates a complete customer journey without any backend dependency.
- A reusable frontend codebase that can later be connected to a real backend, database, and payment gateway with minimal restructuring.
- Practical, portfolio-ready experience in building a production-style, responsive, state-driven web application using JavaScript and localStorage-based persistence.
- Strengthened understanding of client-side routing, component-style UI rendering, and real-world frontend project organization gained through the internship at INFUST PVT LTD.

#### **4.10 Certificates of Completion and Other Communication Proof**

A copy of the Internship Completion Certificate issued by INFUST PVT LTD, signed by Er. Rakesh Kumar, Managing Director, has been attached in Chapter 3 of this report (Internship Completion Certificate) as proof of successful completion of the internship.

## **CHAPTER 5: BIBLIOGRAPHY / REFERENCES**

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